Project Proposal: Hindi–English Cross-lingual Retrieval with Static and Pretrained Embeddings — A Systems Comparison and Failure Analysis

Computational Linguistics 2 | Pranjal Baranwal (2025114004), Apoorv Joshi (2025114014)

Research Questions

RQ1. How do aligned static embeddings (Word2Vec, fastText) compare with pretrained encoders (LaBSE, mE5) at the word, sentence and paragraph level? The systems differ in objective, training data and unit of representation, so we claim a *comparison of systems*, not a clean "static vs. pretrained" effect; mean-pooled MuRIL and pretrained aligned fastText serve as controls. *Pre-stated hypotheses:* (H1) the gap widens with granularity; (H2) static models match or beat pretrained ones on word-level lexicon induction for frequent words, and fastText beats Word2Vec on inflected forms; (H3) word shuffling hurts only the pretrained models.

RQ2 (core). Which properties of Hindi explain where each system fails: inflection, polysemy, Sanskritic vs. Perso-Arabic vs. English-origin vocabulary, transliterated named entities, word order or something else? The centrepiece is a manual failure taxonomy of the best static and the best pretrained system.

Methods

- **Baselines:** BM25 on T2/T3 only; cross-lingual BM25 only via machine-translated queries (opus-mt hi→en).
- **Static:** Word2Vec and fastText [3] (300-d) trained on equal-sized Hindi and English Wikipedia, aligned with Procrustes [9] and retrieved with CSLS [1]; sentence vectors via SIF [5]. Hindi Wikipedia is small, so pretrained fastText wiki/cc vectors serve as a reference point.
- **Pretrained:** LaBSE [2]; mE5 [4] with the required `query:`/`passage:` prefixes; MuRIL [6] (control).
- **Tasks:** (T1) lexicon induction, P@1/5; (T2) parallel sentence retrieval, Acc@1/MRR; (T3) Hindi question to English paragraph retrieval, Recall@k/MRR.
- **Analysis:** inflection and lemma features from Stanza Hindi [11] (UD); frequency and OOV bins; vocabulary layers via Wiktionary etymology categories; named-entity density; word shuffling; and a ∼40-sentence polysemy probe (*kal*, *aam*, *sona*) in which a system succeeds if it ranks the sense-matching English sentence above a contrasting one (static models only via context words), reported qualitatively.
- **Plan:** 4 weeks (wk1 baselines and encoders; wk2 static models and T1; wk3 T3 and analyses; wk4 taxonomy and report).

Datasets

MUSE en–hi dictionaries [1] (T1). **FLORES-200** [7] hi/en, ∼2k sentences (T2). **XQuAD** [8] (T3): its ∼240 paragraphs alone would saturate, so we pool them with ∼10k English Wikipedia distractors; **MIRACL-hi** [10] (Mr. TyDi lacks Hindi) provides a larger Hindi-side check. **Wikipedia** dumps are used for static training. **Limitation:** LaBSE is trained on translation pairs and mE5 on QA/retrieval data, so FLORES and XQuAD may be in-distribution for them.

Key References

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[6] Khanuja et al. MuRIL. *arXiv:2103.10730*, 2021.
[7] NLLB Team. No Language Left Behind (FLORES-200). *arXiv:2207.04672*, 2022.
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